



NEUROMORPHIC ALGORITHMS FOR SCIENTIFIC MACHINE LEARNING A CO-DESIGN STORY

Brad Theilman¹, Ruyin Wan², Qian Zhang², Sanjith Thirunavukkarasu², Sangjae Park⁴, Philip Stachura⁴,
Yu Kevin Cao⁴, Zhenman Fang⁴, Eric C. Cyr¹, Panos Stinis³, George Em Karniadakis²

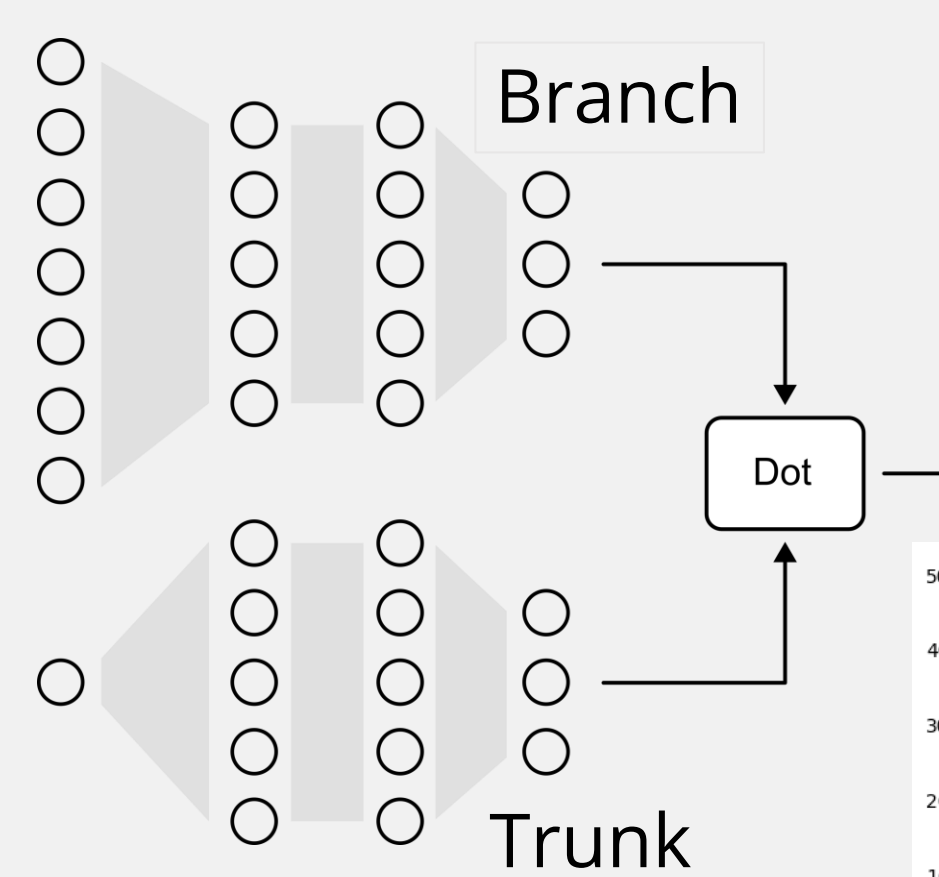
¹Sandia National Laboratories; ²Brown University; ³Pacific Northwest National Laboratory; ⁴University of Minnesota

Introduction

- Neuromorphic hardware can significantly **improve computational efficiency**.
- Spiking neural networks (SNNs) are event-driven NN that naturally **map to efficient execution** on neuromorphic hardware.
- We chart a path to neuromorphic SciML solutions balancing **numerical accuracy** and **low-precision hardware**.

1) How do we represent a SciML model as an SNN?

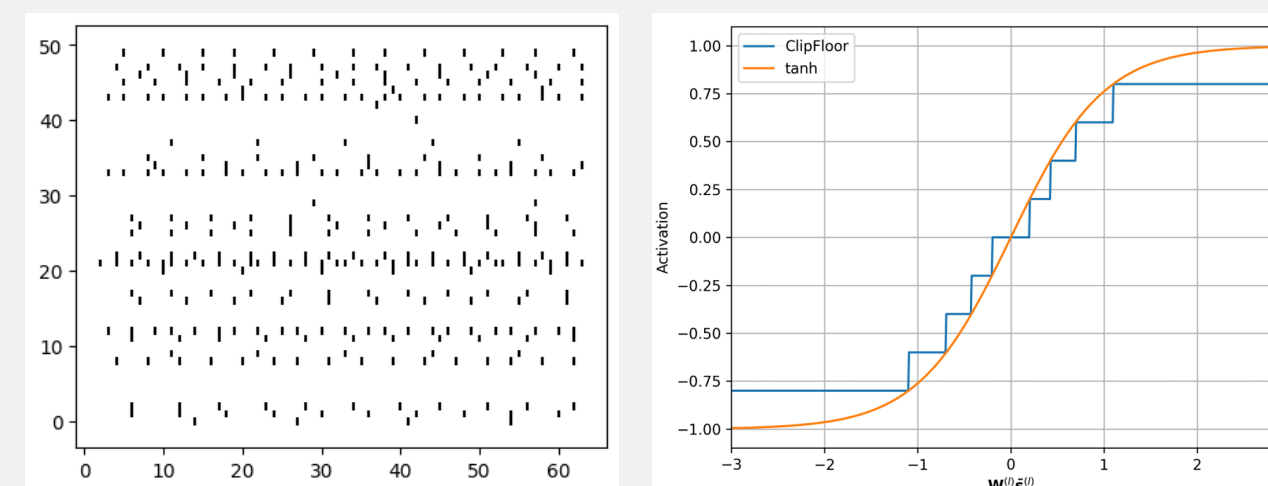
DeepONet Architecture:



LIF Equations:

$$V_i(t+1) = V_i(t) + \sum_j W_{ij} s_j(t) + b_i$$

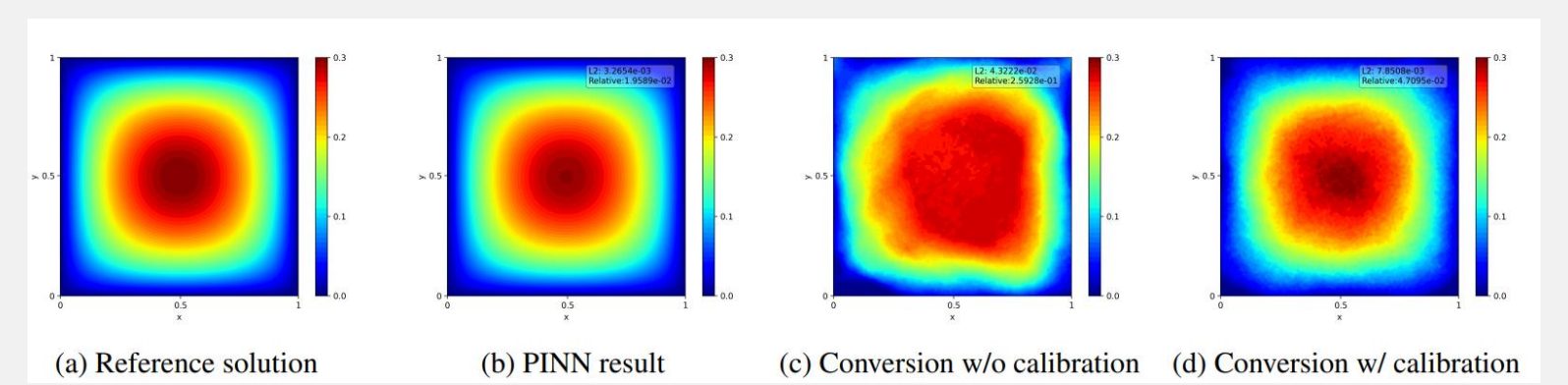
$$s_i(t) = \begin{cases} V_+, & V_i(t) > V_+ \\ 0, & V_- \leq V_i(t) \leq V_+ \\ V_-, & V_i(t) < V_- \end{cases}$$



SNN Conversion Error Bound:

$$\mathbf{e}^{(n),T} \mathbf{H}^{(n)} \mathbf{e}^{(n)} \leq \sum_{l=1}^n 2^{n-l+1} \mathbf{e}_c^{(l),T} (\mathbf{H}^{(l)} + K_L^{(l)} \sqrt{L} \mathbf{I}) \mathbf{e}_c^{(l)}$$

SNN calibration results:



We developed an SNN calibration algorithm that minimizes the error introduced by spiking conversion

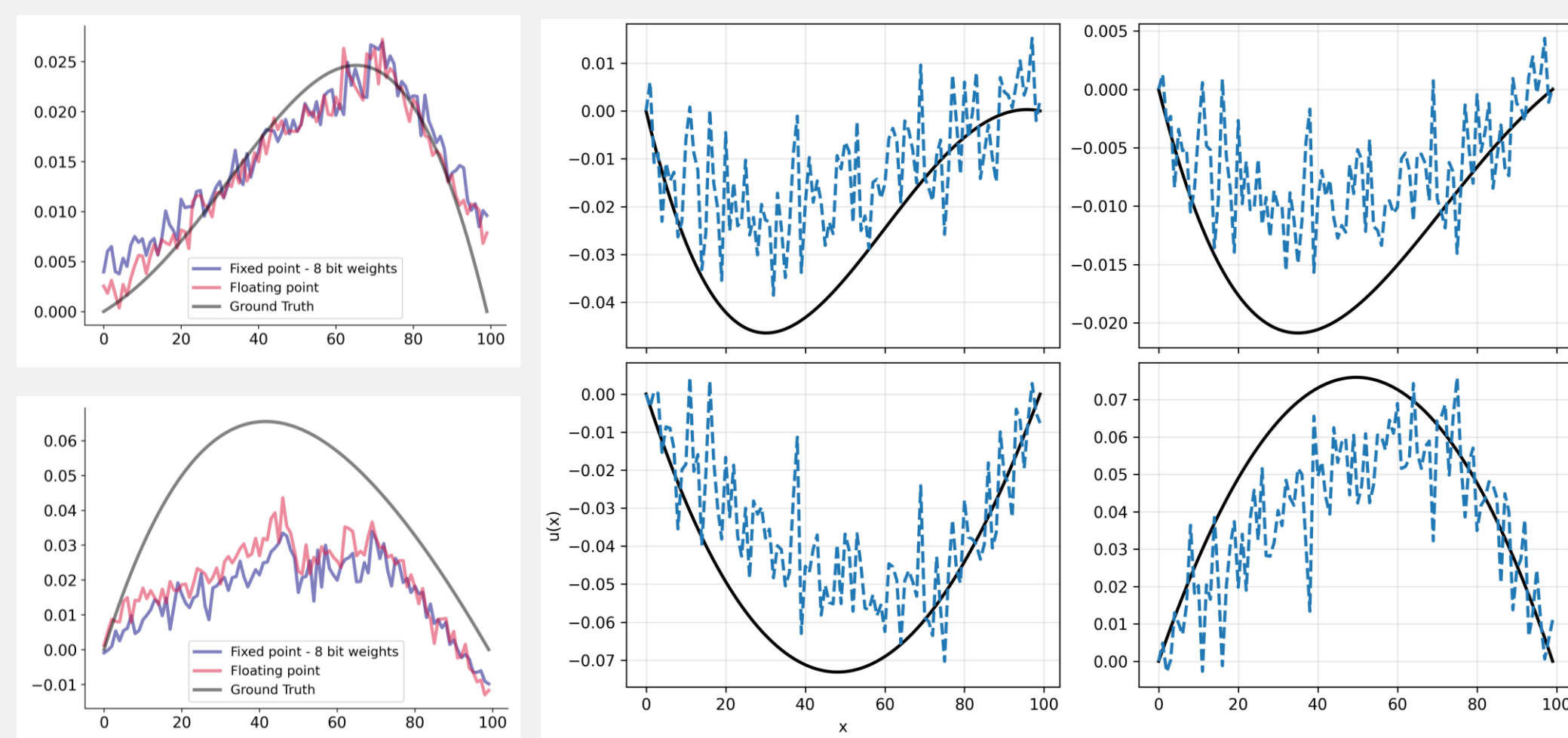
2) Fixed-Point Conversion for SNNs

Our conversion algorithm

- Reparametrizes SNN
- Preserves neuron dynamics
- Uses full fixed-point dynamic range

$$W'_{i+1,i} = \frac{a_i}{a_{i+1}} W_{i+1,i}$$

$$b'_{i+1} = \frac{W_{i+1,i} c_i + b_{i+1}}{a_{i+1}}$$



We tested our converted models on Intel's Loihi 2 chip, but we found an execution incompatibility that prevented realizing high-accurate solutions with our SNNs

3) Quadratic Integrate and Fire (QIF) Neurons

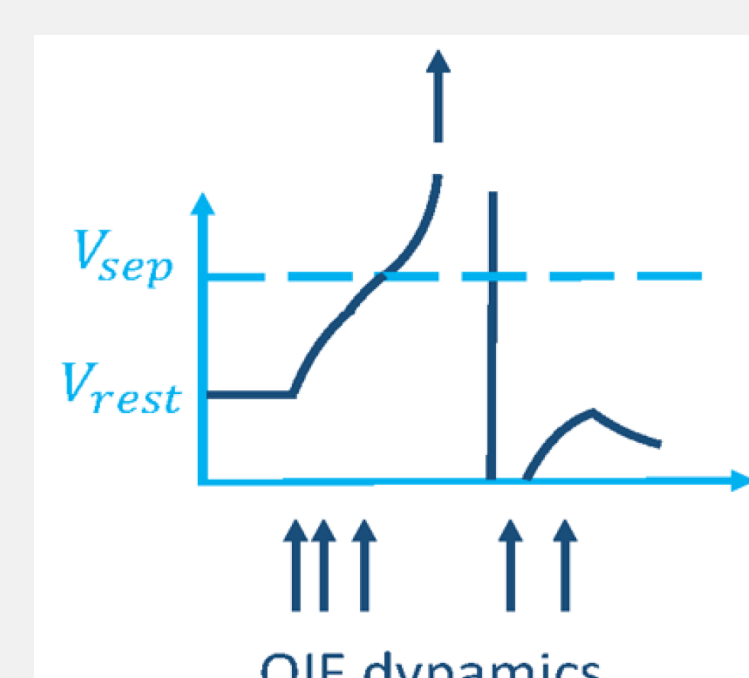
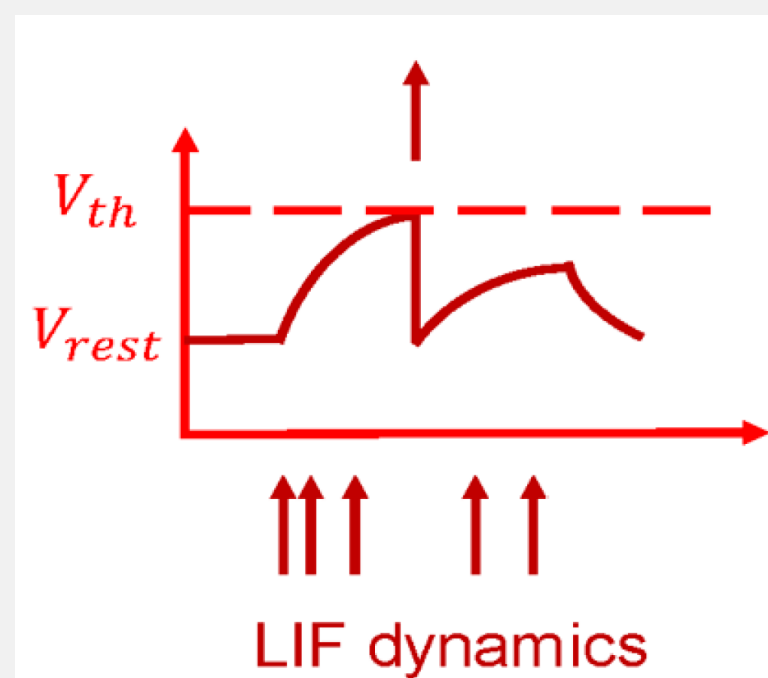
Quadratic integrate and fire (QIF) neurons

- have a nonlinear update rule
- spike times are **differentiable** functions of input
- accurate training** with gradient descent.

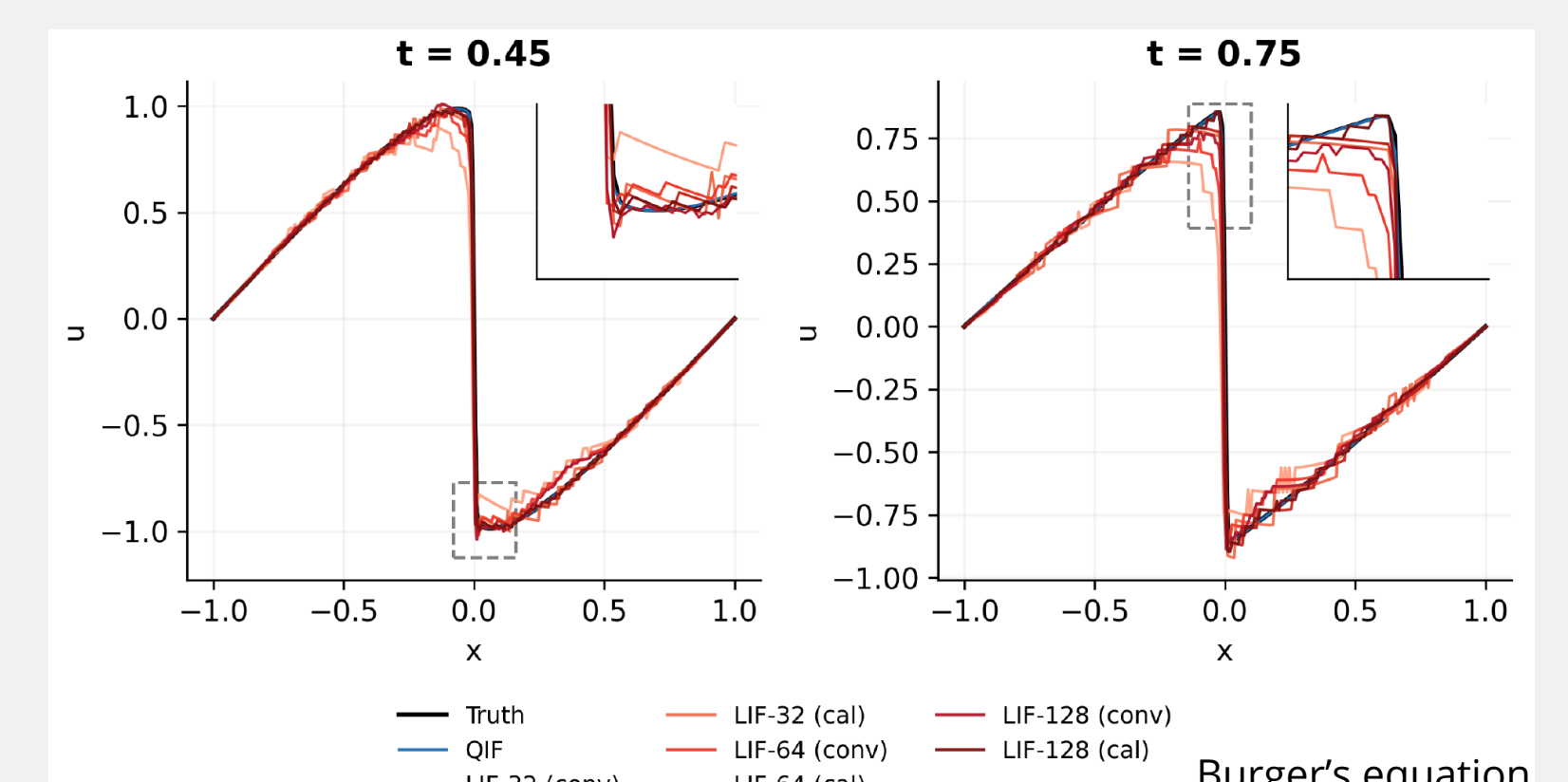
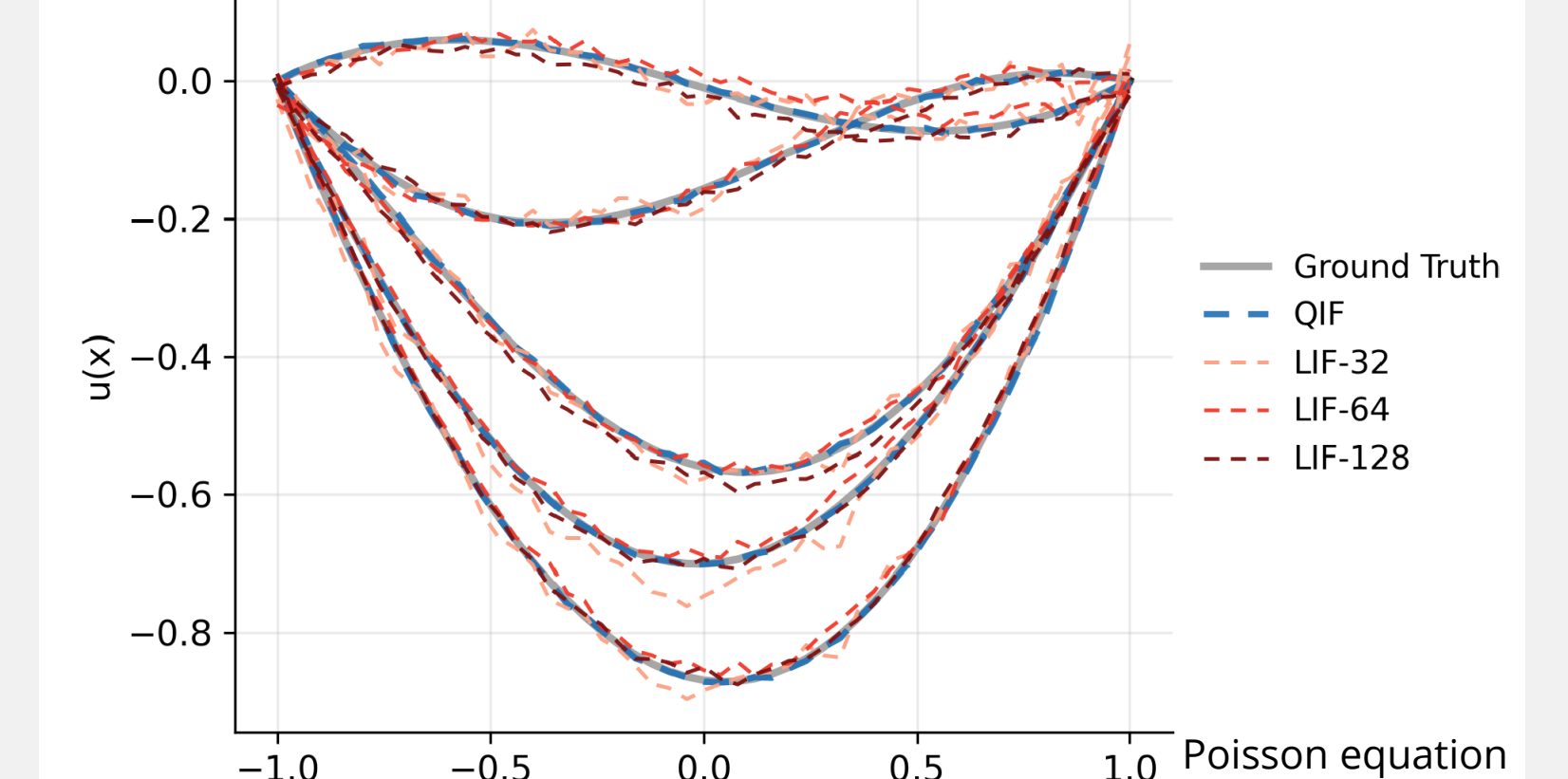
Improves accuracy of spiking SciML models

$$\dot{V}(t) = V(t)(V(t) - 1) + I(t)$$

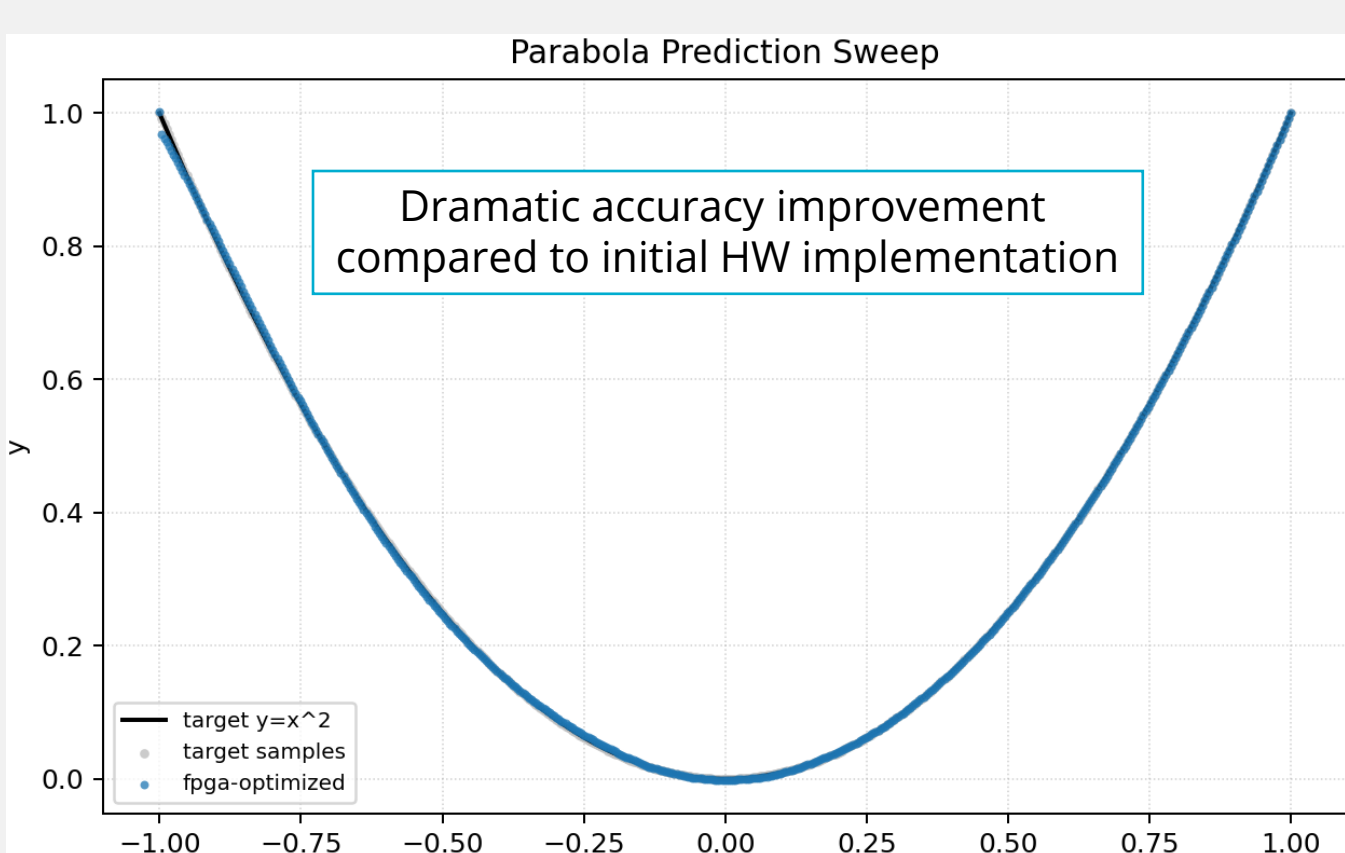
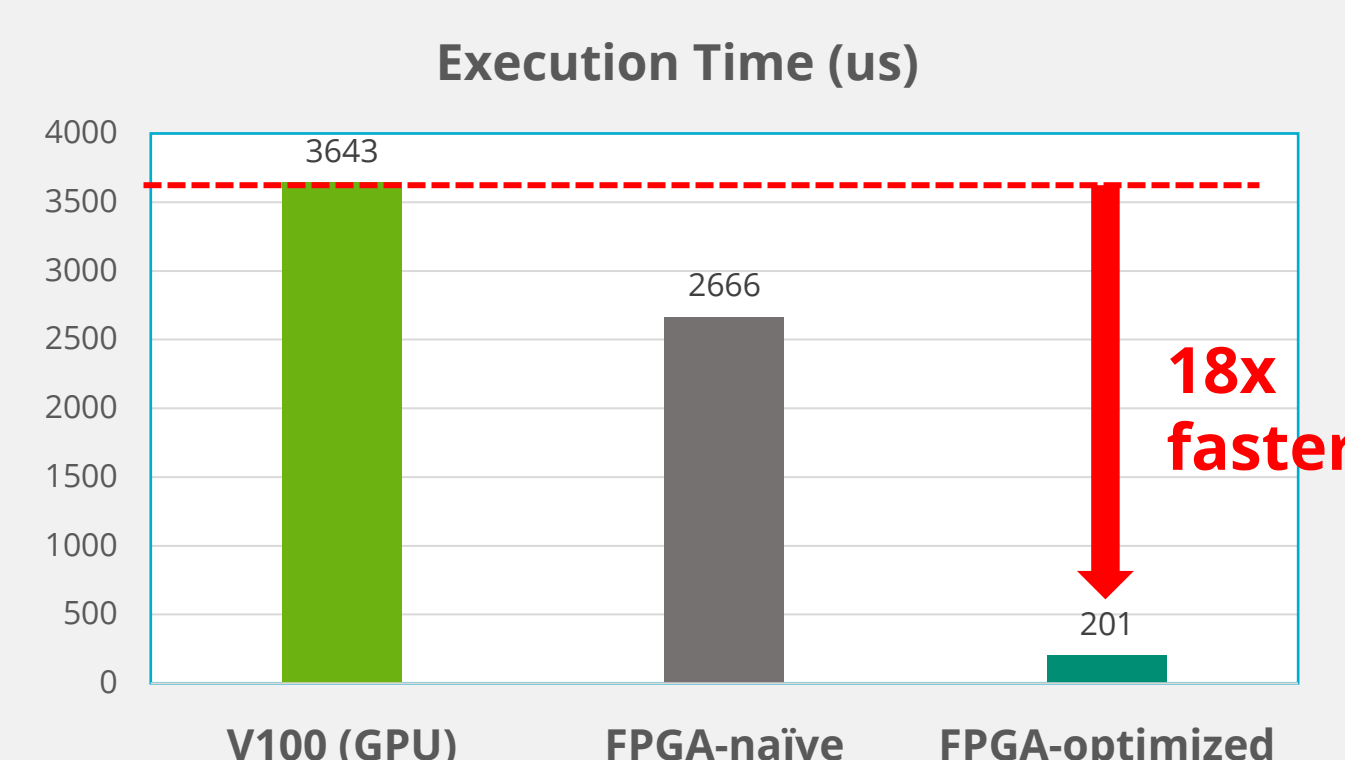
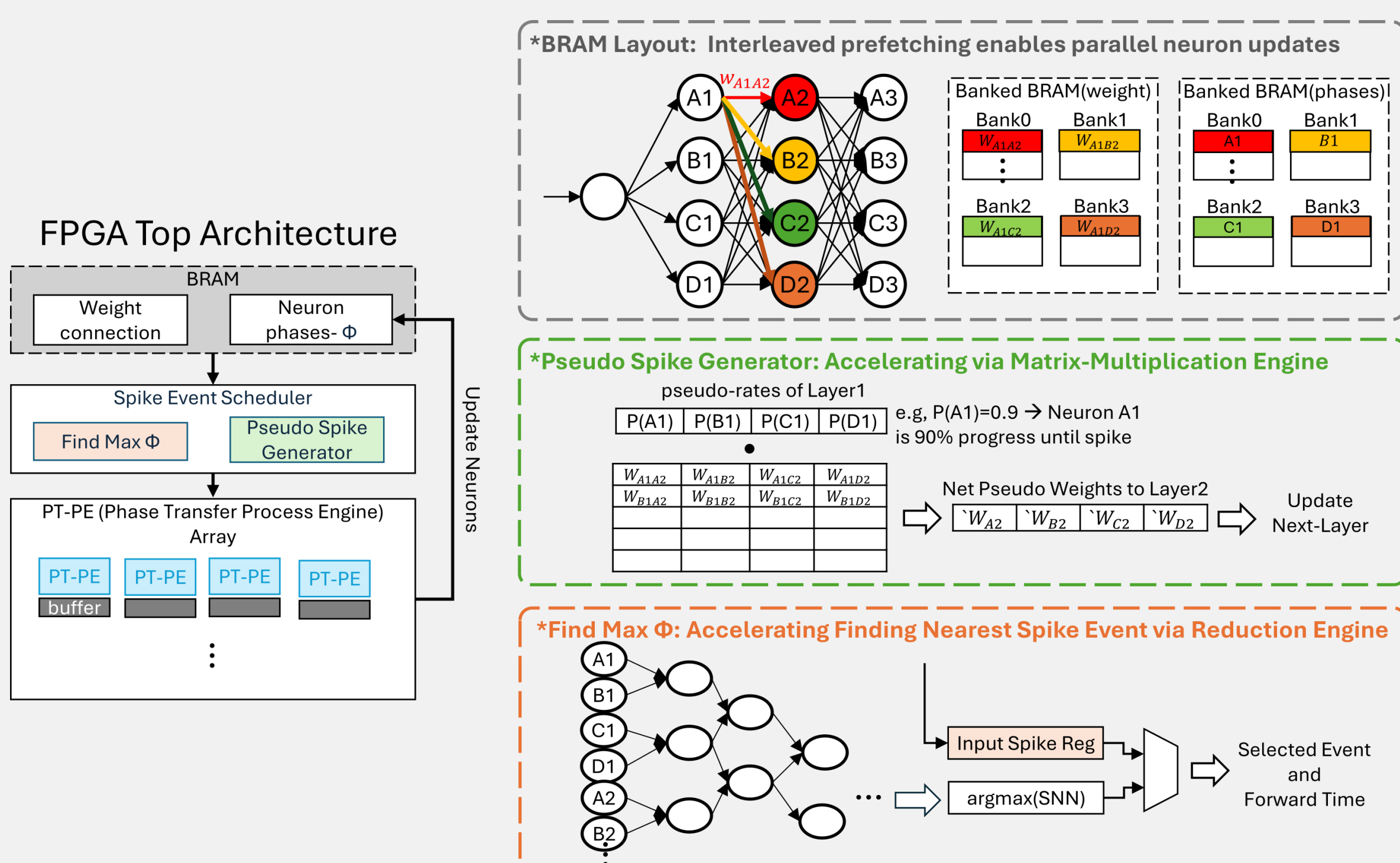
$$I(t) = I_0 + \tau_m \sum_i w_i \sum_{t_i} \delta(t - t_i)$$



DeepONet Poisson: 5 Sample Curves (dashed = predictions)



4) Hardware implementation of QIF



Accuracy

Method	Max abs_error	Mean_abs_error	RMSE
FPGA-naïve	0.0063	0.0017	0.0024
FPGA-optimized	0.0063	0.0017	0.0024

Resource Utilization

Method	LUT	FF	BRAM	DSP
FPGA-naïve	20,543 (7.49%)	26,095 (4.75%)	22.5 (2.46%)	35 (1.38%)
FPGA-optimized	172,033 (62.77%)	166,932 (30.45%)	110.5 (12.12%)	723 (28.69%)

References

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